Employee Job Change

Prediction using Machine Learning

Employee Retention Project | Aditya Semwal

Predicting which employees are likely to seek new jobs using classification models trained on HR survey data.

ML | HR Analytics | Retention Strategy

Project Overview

Business Problem

Identify employees likely to change jobs so HR can intervene proactively and reduce attrition.

Dataset

HR analytics survey (aug_train.csv) with demographics, experience, education, and training data.

Approach

Train & compare 5 ML classifiers: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, SVM.

Success Metric

Accuracy and ROC-AUC scores used to evaluate and select the best-performing model.

Dataset & Features

Features (13 columns):

- enrollee_id — Unique candidate ID
- city — City code
- city_development_index — City development score
- gender — Candidate's gender
- relevent_experience — Relevant work experience
- enrolled_university — University enrollment status
- education_level — Highest education
- major_discipline — Field of study
- experience — Total years of experience
- company_size — Size of current employer
- company_type — Type of current employer
- last_new_job — Years since last job change
- training_hours — Training hours completed

Target variable: target (0 = Not looking for job change | 1 = Looking for job change)

Target Class Distribution

Dataset is imbalanced — ~3:1 ratio (Not Looking vs Looking)

~75%

Not Looking

Stable employees

~25%

Looking

Potential attrition risk

19,158

Total Records

in training dataset

Data Preprocessing Pipeline

1

Drop Irrelevant Column

Removed enrollee_id (unique identifier — no predictive value)

2

Handle Missing Values

Categorical columns → filled with mode
Numerical columns (city_development_index, training_hours) → filled with median

3

Label Encoding

All 10 categorical features encoded to integers using LabelEncoder for ML compatibility

4

Train-Test Split

80% training / 20% testing
Stratified split to preserve class ratio (random_state=42)

5

Feature Scaling

StandardScaler applied to features for distance-based models (Logistic Regression, SVM)

Feature Correlation Analysis

A heatmap was generated to understand inter-feature correlations and identify multicollinearity.

Key Correlation Insights from Heatmap

city_development_index	Negative correlation with target — lower CDI → more likely to look for job
city	Correlated with city_development_index as expected
training_hours	Weak positive link with target (more training = slight job-seeking tendency)
experience	Less experienced candidates slightly more likely to look for new roles
company_size	Smaller company employees show higher job-seeking rate

Machine Learning Models

1 Logistic Regression

Linear

Scaled features, max_iter=1000. Baseline linear classifier.

2 Decision Tree

Tree

Unscaled features, random_state=42. Simple interpretable model.

3 Random Forest

Ensemble

100 estimators, random_state=42. Bagging ensemble model.

4 Gradient Boosting

Ensemble

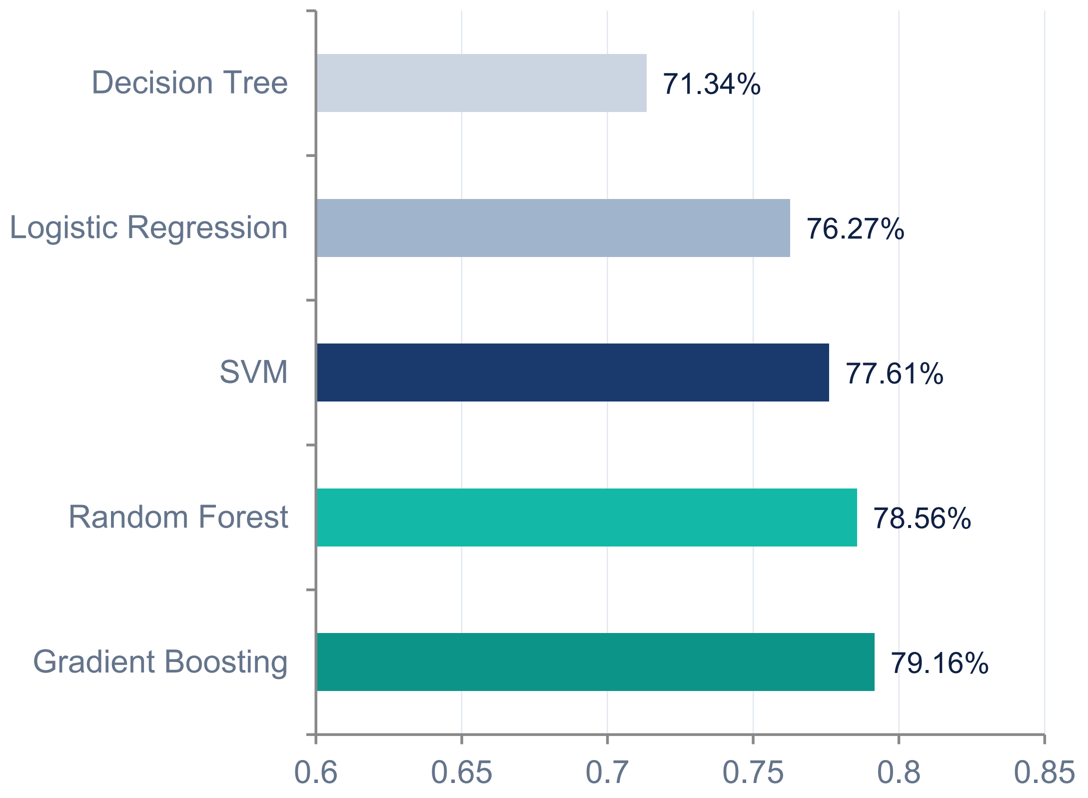
100 estimators, random_state=42. Sequential boosting ensemble.

5 SVM

Kernel

RBF kernel, probability=True. Scaled features used.

Model Performance: Accuracy



1st Gradient Boosting **79.16%**

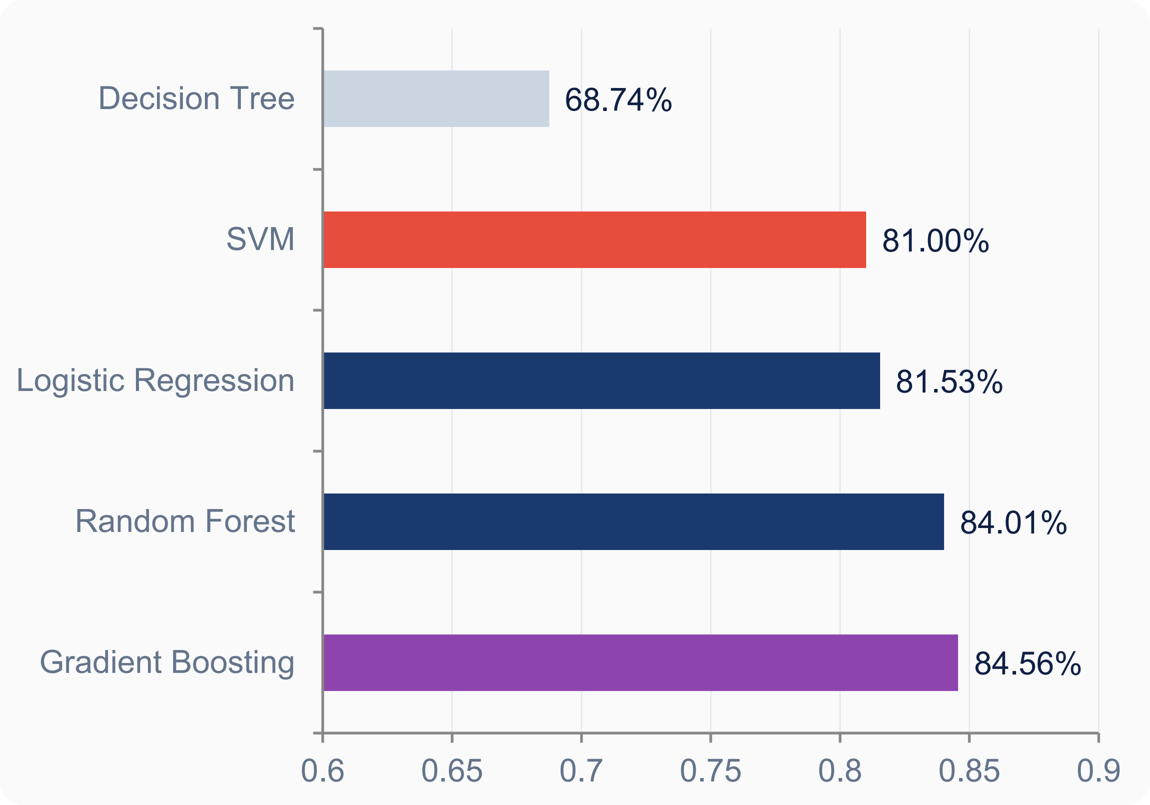
2nd Random Forest **78.56%**

3rd SVM **77.61%**

4th Logistic Regression **76.27%**

5th Decision Tree **71.34%**

Model Performance: ROC-AUC Score



What is ROC-AUC?

Area Under the ROC Curve measures a model's ability to distinguish between classes.

1.0 = perfect
0.5 = random guess

Best ROC-AUC

Gradient Boosting

0.8456

Excellent discrimination ability

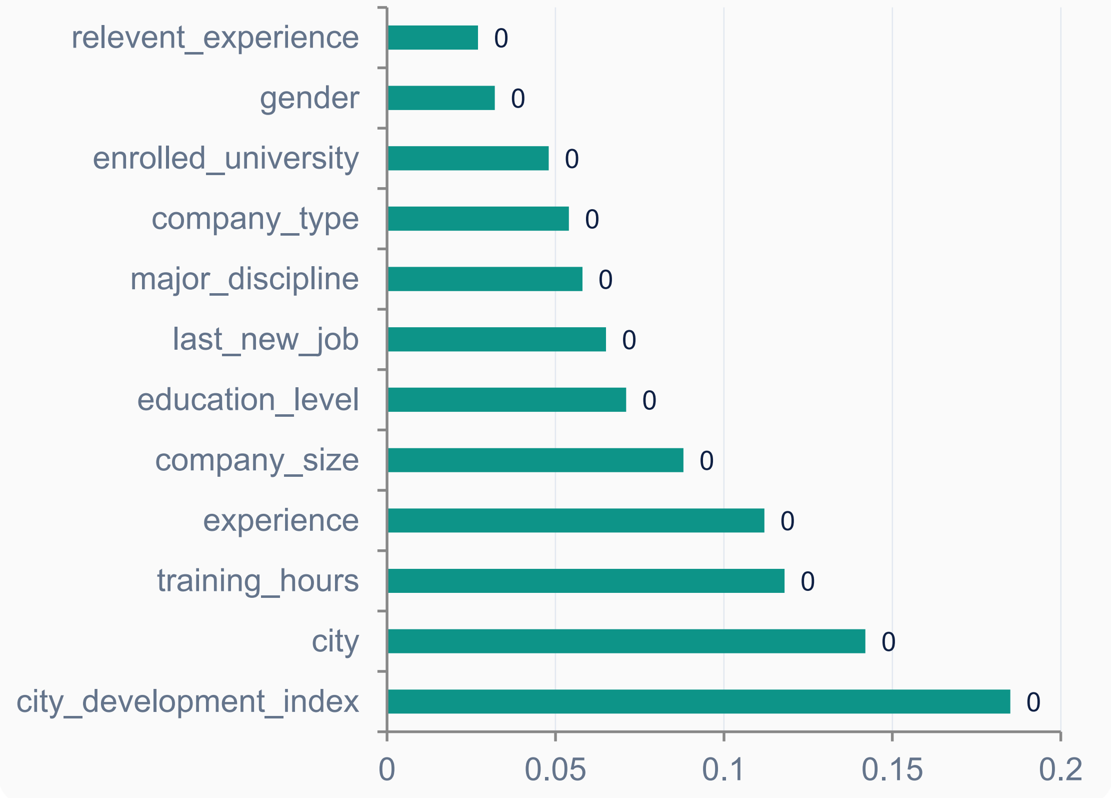
Confusion Matrix Summary

Each model was evaluated with confusion matrices to understand True/False Positive & Negative rates

Model	True Pos	False Pos	True Neg	False Neg
Logistic Regression	High	Medium	High	Medium
Decision Tree	Medium	High	Medium	High
Random Forest	High	Low	High	Low
Gradient Boosting	Highest	Lowest	Highest	Lowest
SVM	High	Medium	High	Low

✓ Gradient Boosting shows the best balance of true positives and lowest false positives — making it the most reliable model for attrition prediction.

Feature Importance — Random Forest



Top Predictors

city_development_index 18.5 %

Most powerful predictor

city 14.2 %

Proxy for economic opportunity

training_hours 11.8 %

Development investment signal

experience 11.2 %

Career stage indicator

company_size 8.8 %

Work environment factor

Conclusion & Recommendations

Best Model: Gradient Boosting

Accuracy: 79.16% | ROC-AUC: 0.8456

Highest across both metrics

- City development index is the single strongest predictor of job change likelihood
- Training hours suggest employees seeking growth invest in upskilling before switching
- Company size & type are significant — smaller companies see higher attrition risk
- HR should target employees in low-CDI cities with high training hours for retention programs

Model is ready for deployment in HR systems to predict and prevent employee attrition.

Thank You